

Our final exam will be held on Friday, May 9th, 8:30 - 11am in our usual classroom, Hodson 113.

The final exam is cumulative, meaning that it will test material from all parts of the course. However, you can expect to see more of an emphasis on material that we have covered since the second midterm exam, since that material has not yet been tested.

Exams will be assigned on paper and you will not be asked to use the computer during the test.

You may use one double-sided 8.5 x 11in sheet of notes. You may use a calculator, but not your phone. If you need to borrow a calculator, please ask. No collaboration is allowed as this exam should assess your personal understanding so that I can properly award you credit for your learning.

To study for this test, I strongly recommend that you pull out your old homework assignments, labs, and quizzes, hide the answers from yourself and re-do the work, checking your answers against the solutions that are posted on Blackboard. The exercise sections in our textbook are another good source of problems. Remember that answers to odd-numbered exercises are at the “back” of the book.

Things to know and be able to do

In all questions on the exam, your ability to explain and interpret the meaning of your calculations will carry quite a bit of weight. As you are studying, make sure to review how to write paragraphs that describe distributions, how to write sentences of interpretation for hypothesis testing - where we are comparing experimental results to a null model, how to write sentences of interpretation for coefficients of a linear regression model, how to write sentences of interpretation for confidence intervals, etc.

Things to know and be able to do

Here is a list of things to know and be able to do, listed in reverse order: material we have covered since Exam 2, Exam 2 material, Exam 1 material.

Since Exam 2

Here is a brief summary of what we have covered since Exam 2.

Data basics, study design, describing and summarizing single variables

Though we first learned these concepts before Exam 1, we have continued to emphasize these skills since Exam 2.

- Be able to identify cases, variables (numerical or categorical) in a study.

- Be able to identify the population, sample, parameters, statistics, explanatory (or predictor) variables, response variables in a study.
- Be able to comment on whether a particular study lends itself to generalization and / or to drawing causal conclusions.
- Numerical variables
 - Given a visualization of the distribution of a single numerical variable, be able to write a description of the distribution, first writing a who/what/where/when sentence, and then continuing the paragraph addressing center, shape, spread, outliers, and any other interesting features. All such descriptions should be in context, and should be supported with numbers from the graph or from computed statistics.
 - Be able to use appropriate measures of center (mean or median) and spread (standard deviation or quantiles (middle 50% and middle 95%)). Remember that you should use different combinations of these depending on whether your distribution is symmetric or skewed. Mean + standard deviation are good for symmetrical distributions. Median + quantiles are good for skewed distributions.
 - Be able to say which sort of graph would be appropriate for a particular visualization.
- Categorical variables
 - Given a visualization of a categorical variable, or a pair of categorical variables, be able to write a paragraph highlighting key features of the distribution, speaking in context, and supporting your points with numbers drawn from the graph.
 - Given a contingency table, be able to compute row and column proportions to back up claims you make about a study.
 - Be able to comment on whether there is an association between two categorical variables, given a visualization or a contingency table for these variables.

New since Exam 2

- **Sampling distributions:** Be able to clearly describe how to create a sampling distribution for a statistic and be able to state its properties for shape, center, spread. Know how sample size affects the distribution. Why is this important?
- **Bootstrap distributions:** Be able to clearly describe how to create a bootstrap distribution and be able to state its properties for shape, center, spread.
- **Confidence intervals:** Be able to compute confidence intervals for a single proportion, a single mean, a difference of proportions, and a difference of means in several different ways. See the additional Confidence Intervals Summary document, posted with this information sheet.
 - Using the **bootstrapping applets in StatKey**. Create a bootstrap distribution and interpret it.

- Using **an estimated standard error from a bootstrap distribution** and a formula.
 - **Using formulas** such as $statistic +$ or $- 2SE$ and $statistic +$ or $-(2/3)SE$ along with estimated standard errors for calculating the estimated SE using the appropriate formula. Remember, there are different formulas for a standard error for proportions, a standard error for means, and a standard error for a difference. **See the extra handout containing a summary of formulas we use for confidence intervals.**
 - Know when to use z^* or t^* , and how to figure out *which* value of either one to use.
- Be able to carefully write accurate **sentences of interpretation for confidence intervals**.
 - Be able to explain exactly what “95% confidence” means
 - Be able to clearly describe the reason for using a t-distribution when calculating a confidence interval for means.

Exam 2 material

Linear regression - numerical explanatory and numerical response

- Given a research question having a numerical explanatory (predictor) and numerical response, know that a scatterplot is the appropriate visualization.
- Be able to describe the relationship between two numerical variables, given a scatterplot depicting the relationship. Descriptions should address shape, strength, and direction of the association, should use the “General trends, Examples, Exceptions” (GEE) model for description, and should be in context. Examples illustrating the direction of the association can include using the “dividing the graph into thirds” method for description or the direction can be illustrated by using the sentence for interpreting the slope of the regression line. Examples illustrating the strength of the association should provide examples of the variability of the response variable for given values of the explanatory variable, as we have done in class and in homework solutions.
- Be able to use the R model summary output to write the regression equation.
- Be able to write a sentence of interpretation (in context) for the slope of a regression line. I’m looking for a sentence of very specific format here, as we practiced in class and on homework.
- Be able to write a sentence of interpretation (in context) for the intercept of a regression line. I’m looking for a sentence of very specific format here, as we practiced in class and on homework.
- Be able to use the equation of a regression line to find a predicted value of the explanatory variable. Care must be taken to include units, for full credit.

- Be able to calculate the residual of a data point, showing all work. Care must be taken to include units for full credit.
- Just by looking at a data point on a scatterplot where the relationship has been modeled using a line, be able to say whether the residual will be positive or negative.

Linear regression – categorical predictors and multiple predictors

- Given regression model summary output involving a single categorical predictor and a numerical response, be able to write the regression equation.
- Given regression model summary output involving a single categorical predictor and a numerical response, be able to write sentences of interpretation for the coefficients of the model, in context, and in the style we have practiced in class and on homework.
- Given regression model summary output involving a categorical predictor and a numerical predictor for a numerical response variable, be able to write the regression equation.
- Given regression model summary output involving a categorical predictor and a numerical predictor for a numerical response variable, be able to write sentences of interpretation for the coefficients of the model, in context, and in the style we have practiced in class and on homework.

Learning from experiments: comparing effects of treatments on groups

- Given the description of a study with control and treatment groups, be able to analyze and answer the question of whether it seems plausible that the observed difference between the treatment groups is due to the natural variability associated with the random assignment of subjects to treatment groups. (E.g. Sex Discrimination study, Weight Loss study). More specifically, you should be able to:
 - Describe how to perform a simulation using playing cards or technology to investigate this question.
 - For a given dotplot for such a simulation, be able to tell exactly how many dots are on the dotplot and what each dot represents.
 - Write null and alternate hypotheses, both in words and with the correct notation.
 - Be able to write a paragraph describing the dotplot, in context.
 - Be able to estimate and report a one-sided p-value.
 - Be able to write a sentence telling how the p-value influences your answer to the question of compatibility of the observed difference with the null model.

Learning from experiments: assessment of the null model in a study involving one proportion or one mean

- Given the description of a study in which the statistic under consideration is a single proportion or mean, be able to analyze and answer the question of whether the null

model of no effect provides a plausible explanation for the observed proportion or mean. (E.g. Helper/Hinderer study, ESP study). More specifically,

- For a given dotplot for a randomization test, be able to tell exactly how many dots are on the dotplot and what each dotplot represents.
- Write null and alternate hypotheses, both in words and with the correct notation.
- Be able to write a paragraph describing the dotplot, in context.
- Be able to compute and report a one-sided p-value.
- Be able to write a sentence telling how the p-value influences your answer to the question of compatibility of the observed proportion with the null model.

Exam 1 material

Data basics

- Be able to identify the cases and variables in a dataset and identify variables as numerical or categorical.
- Be able to word variables carefully and appropriately for the cases in a dataset.
- Be able to create a hypothetical first few rows of a dataset for a given study in a “tidy data” manner (cases in rows, variables in columns).
- Be able to pose an appropriate research question about one variable or about more than one variable in a dataset.

Study design: Generalization

- Be able to identify the population and the sample in a study.
- Be able to identify parameters and statistics from a study.
- Be able to discuss several different types of random sampling methods. Make sure you are especially able to explain how to select a simple random sample.
- Be able to explain why random sampling is so important.
- Be able to recognize sampling bias and explain why it is necessary to avoid sampling bias.
- Given a particular study, be able to discuss whether the study lends itself to generalization. (We have had several homework and quiz questions along these lines.)

Study design: Causation

- Be able to identify variables as explanatory or response and explain how you can tell.
- Be able to identify a study as an experiment or an observational study and how you can tell.
- Be able to comment carefully on whether a study can be used to establish causal relationships.
- Be able to propose a possible confounding variable, and explain its connection to the explanatory and response variables in a way that gives an alternative explanation for the connection between the explanatory and response variables.
- Be able to identify a study as being blind or double-blind and explain the importance of this.

Study design: Big picture

- Be able to explain how you would design a study from start to finish so that it lends itself to generalization to the population of interest and to establishing causal relationships. (Hint: Surveys are no good here! Why not?)
- Given the description of a study, be able to critically assess its worth, point out potential flaws, and suggest improvements in the design.

Describing numerical variables

- Given a visualization of the distribution of a single numerical variable, be able to write a description of the distribution, first writing a who/what/where/when sentence, and then continuing the paragraph addressing center, shape, spread, outliers, and any other interesting features. All such descriptions should be in context, and should be supported with numbers from the graph or from computed statistics.
- Be able to compare the distributions of a numerical variable grouped by a categorical variable. Comparisons should address the characteristics outlined in the above bullet point.
- Be able to use appropriate measures of center (mean or median) and spread (standard deviation or quantiles (middle 50% and middle 95%)). Remember that you should use different combinations of these depending on whether your distribution is symmetric or skewed. What are those combinations?
- Be able to explain what a robust (or resistant) statistic is and give examples.

Describing categorical variables

- Given a visualization of a categorical variable, be able to write a paragraph highlighting key features of the distribution, speaking in context, and supporting your points with numbers drawn from the graph.

- Given a contingency table, be able to compute row and column proportions to back up claims you make about a study.
- Be able to comment on whether there is an association between two categorical variables, given a visualization or a contingency table for these variables.